T. M. Mehrab Hasan

Bachelor of Science in
IoT & Robotics Engineering

Skills

- Robotics & Embedded System: Embedded systems, Programming of Microcontrollers (Arduino, ESP8266), Sensors & Actuators, Wireless Embedded Systems.
- Programming language: C, C++, Python, Django, HTML, CSS, PHP, Javascript, React, Dart, NextJS, NodeJS.
- Database: MSSQL, MySQL, SQLite, Firebase: Firestore, FirebaseAuth, Firestorage, RealTime database, MongoDB, PostgreSQL, RDBMS.
- Operating System: Windows, Linux.
- Application Software: MS office Applications, LaTeX, Kali Linux, Arduino IDE, Netbeans IDE, Anaconda, VS Code.

Language

- Bengali: Mother tongue
- English: Proficient user (C1)
  IELTS Score: 7.0 (Overall)

Career Objective

With a strong foundation in IoT and Robotics Engineering and growing expertise in AI, Machine Learning, and Software Development, I strive to engineer intelligent and connected systems that solve real-world challenges. As an enthusiast of AI/ML Engineer, and Software Engineer, I focus on integrating advanced sensing technologies with responsive web and software platforms to improve efficiency, automation, and user experience. Driven by continuous learning and creative problem-solving, I aim to contribute impactful, technology-driven solutions to industry and society.

Education

B.Sc. Engg. (IRE) — 2025
Department of IoT & Robotics Engineering
University of Frontier Technology, Bangladesh
Major: IoT & Robotics Engineering
CGPA: 3.71/4.00

Higher Secondary Certificate (HSC) — 2019
Notre Dame College, Dhaka
Dhaka Board, Bangladesh
Major: Science
GPA: 5.00/5.00

Secondary School Certificate (SSC) — 2017
Monipur High School and College
Dhaka Board, Bangladesh
Major: Science
GPA: 5.00/5.00

Projects

- BITSS VWAR Software (Portable Version):
  After the completion of the windows version of BITSS VWAR Software, currently the portable version is under development. This software is being developed at BFIN IT PVT. LTD
- BITSS VWAR Software (Windows Version):
  BITSS VWAR Software (Windows) provides device security; filtering of malware; identifies, neutralises and removes malware implanted in the device that supports both offline & online mode. Currently, has already completed the full production ready BITSS VWAR Software (Windows Version) to be launched in the market. This software was developed at BFIN IT PVT. LTD.
- BAAZAR X - AI-Powered E-Commerce Platform:
  BAAZAR X is a modern, AI-powered e-commerce platform featuring a clean white design with neon blue accents and golden highlights. The platform includes comprehensive product management, intelligent chatbot assistance, and a complete admin dashboard for managing the store. this project was completed during my work at PERPEX.
- Real-Time Water Quality Monitor Using ESP32 with Local Server & Firebase:
  An IoT-based smart water quality monitoring system using an ESP32 and multiple analog/ digital sensors to measure pH, TDS, turbidity, and temperature. The sensor data is uploaded to Firebase Realtime Database and visualized on a live Python Flask dashboard. This project was completed during my working at Robo Tech Valley.
- Intelligent Investment Advisor: A Stacked ML Based Recommendation System with Explainable Insights:
  Developed a Stacked ML-based system for investment recommendations and

Contact

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TM-Mehrab-Hasan

T. M. Mehrab Hasan

Mehrab Hasan

Orcid / Mehrab Hasan

Professional Experience

- Jr. Software Developer
  (August 2025 - Current)
  BFIN IT PVT. LTD., Bangladesh
- Django Backend Developer (Remote)
  (September 2025 - Current)
  NSR DEV, Bangladesh
- Django (Python) Intern
  (July 2025 - September 2025)
  Perpex, India
- Robotics & Embedded System Engineer
  (May 2025 - July 2025)
  Robo Tech Valley, Bangladesh
- Lead Web Developer
  (Jan 2022 - Feb 2023)
  Greenity Foundation, Bangladesh

Extra Curricular Activities

- Deputy Organizing Secretary
  (May 2023 - December 2024)
  UFTB Robotics Club
- Korean Language Secretary
  (May 2023 - December 2024)
  UFTB Language Club
- Program Secretary
  (January 2023 - December 2024)
  UFTB Cultural Club

provided clear reasoning with Explainable AI. This project was my final year thesis project at University of Frontier Technology, Bangladesh. The research paper of this project has been submitted to "Statistics, Optimization & Information Computing" under IAPress Journal.

- A Smart IoT Solution for Managing Drainage Congestion in Bangladesh:
  Designed an IoT-based system for garbage detection using ML with automated water-flow control, and real-time monitoring with an alert system. This project was built and completed as my capstone project in University of Frontier Technology, Bangladesh. Also the research paper of this project was accepted at the "11th International Conference on Networking, Systems and Security".
- Early Stage Fire Detection Robot:
  Built a mini automated robot that performs movements based on object detection method and identifies any fire near it and also provide automated water sprinkler as first aid. And finally, sends a notification to the owner of fire remotely. The robot is also controllable wirelessly. It was featured at 57th International Education Expo 2023. This project was completed as my Modelling & Simulations course project.

Achievements & Certificates

- Achieved the badge of "Module 1: Personal & Professional Development" from Aspire Institute Leadership Program on September, 2025.
- Recognized for outstanding academic performance and earned the "Dean's Award" on June 29. 2025.
- Achieved "Introduction to Critical Infrastructure Protection" certificate from OPSWAT Academy on February 11, 2025.
- Achieved "Advanced Cyber Security Program" certificate from Team Matrix Elite Hackers on March 15, 2025.
- Became Winner of Applink Innovative Idea Contest with the idea of "Intelligent Law Advisor App" from Applink on February, 2024.
- Earned "21st Century Employability Skilling (Advanced)" certificate from Wadhwani Foundation on December 06, 2023.
- Achieved "Cyber Security Training" certificate from Creative IT Institute on October 18, 2023.
- Attained "21st Century Core Employability Skills Development" certificate from Wadhwani Foundation on October 17, 2023.
- Achieved "Introduction to Cloud" certificate from AWS Academy on August 16, 2025.
- Placed 3rd Runner UP in Upazila Science Fair, Gazipur on May 07, 2023.
- Placed 11th at UFTB Googling Contest hosted by University of Frontier Technology, Bangladesh on December 16, 2022.
- Attained "Basic Python Programming" certificate and secured one of the top positions by CppBuzz on January 25, 2022.

Publications

Published / Accepted

- Cost-Effective High-Reliability IoT-Based Multi-Modal Patient Monitoring with Real-Time Clinical Alerts and Cloud Analytics
  2nd IEEE Conference on Computing Applications and Systems (COMPAS 2025), IEEE, 2025.
  Authors: Sharna Rani Das, Abdul Hasib, T. M. Mehrab Hasan, Saurav Kundu, Arjina Nasrin Khan, Md. Meraj Serker.

- Executive Officer
  (January 2023 - December 2024)
  UFTB Programming Club
- Vice-President (Field)
  & Group representative
  (August 2017 - July 2018)
  Notre Dame English Club
- Executive Member & Volunteer
  (August 2017 - July 2018)
  Notre Dame Science Club

References

- Farzana Akter
  Asst. Professor
  Department of IoT &
  Robotics Engineering,
  University of Frontier Technology,
  Bangladesh.
  E-mail: farzana0001@uftb.ac.bd
  Contact: +880 17415582031

- Md. Habibur Rahman
  Lecturer & Ph.D. (perusing)
  Department of IoT &
  Robotics Engineering,
  University of Frontier Technology,
  Bangladesh.
  E-mail: habibur0001@uftb.ac.bd
  Contact: +1 (346) 353-1143

- A S M Ahsanul Sarkar Akib
  Managing Director
  Robo Tech Valley
  E-mail: akib@robotechvalley.com
  Contact: +880 1781886422

- Edge-to-Cloud Orchestrated Multi-Sensor IoT Framework for Real-Time WQI Classification and Predictive Assessment via Ensemble ML
  2nd IEEE Conference on Computing Applications and Systems (COMPAS 2025), IEEE, 2025.
  Authors: Abdul Hasib, T. M. Mehrab Hasan, Meraj Serker, Md Mursalin Hossain Misat, Jakia Arthy, Arjina Nasrin Khan.

- Simultaneous Detection of LSD and FMD in Cattle Using Ensemble Deep Learning
  International Conference on Big Data, IoT and Machine Learning, Springer, 2025.
  Authors: Nazibul Basar Ayon, Abdul Hasib, Md. Faishal Ahmed, Md. Sadiqur Rahman, Kamrul Islam, T. M. Mehrab Hasan.

- Resilient ESP32 Architecture for Unified Fleet Integrity and Emergency Safety in Resource-Constrained Environments
  International Conference on Big Data, IoT and Machine Learning, Springer, 2025.
  Authors: T. M. Mehrab Hasan, Abdul Hasib, Md. Meraj Serker, Kamrul Islam, Arjina Nasrin Khan, Md. Shakil.

- Design and Development of a Low-Cost Scalable GSM-IoT Smart Pet Feeder with a Remote Mobile Application
  International Conference on Signal Processing, Information, Communication and Systems 2025, IEEE, 2024.
  Authors: Md. Rakibul Hasan Nishat, S. M. Khalid Bin Zahid, Abdul Hasib, T. M. Mehrab Hasan, Mohammad Arman.

- Towards Cleaner Cities: A Smart IoT Solution for Managing Drainage Congestion in Bangladesh
  11th International Conference on Networking, Systems and Security, ICPS.
  Authors: T. M. Mehrab Hasan, Farzana Akter, Mahjabin Ashka.

Under Review

- Intelligent Investment Advisor: A Stacked ML-Based Investment Sector Recommendations with Explainable Insights
  Statistics, Optimization & Information Computing, IAPress, 2025.
  Status: Under review.
  Authors: T. M. Mehrab Hasan, Fahmid Al Farid, Rakib Hossen, Mst Deloara Khushi, Md. Masud Rana, Anichur Rahman, Zobayer Ahmed, Hezerul Abdul Karim, Abu Saleh Musa Miah.

Declaration of Authenticity

I, hereby, declare that all information I have presented so far is true to my knowledge. If required and where applicable, this document can be supported by authentic certificates/papers.

T. M. Mehrab Hasan